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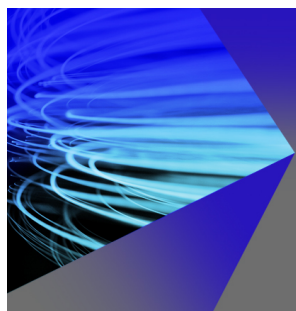
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ABSTRACT

The accurate prediction of pressure drop in packed beds is fundamental to numerous chemical, mechanical, and environmental engineering applications. While traditional models such as the Ergun equation and Reynolds number-based correlations have been widely employed, they suffer from inherent limitations and lack a unified framework bridging packed beds and empty conduits. This paper presents a comprehensive derivation of the Quinn Fluid Flow Model (QFFM), a novel theoretical framework developed from first principles. The QFFM establishes a universal linear relationship between the normalized dimensionless pressure gradient (P_Q) and the fluid current (C_Q), expressed as $P_Q = k_1 + k_2 C_Q$. A key innovation is the conceptualization of any closed conduit as a “packed conduit” via the Hypothetical Q-Channel (HQC) and the wall normalization coefficient (λ), which inherently incorporates the effects of tortuosity, viscous boundary layers, and surface roughness. By unifying the description of flow through packed beds, the new semi-empirical model offers a robust, physics-based alternative to conventional semi-empirical correlations, validated across diverse flow regimes and particle morphologies.

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I. INTRODUCTION

The accurate prediction of pressure drop in fluid flow systems is a cornerstone of chemical, mechanical, and environmental engineering.^{1,2} This is particularly critical for flows through packed beds, which are ubiquitous in industrial applications such as chemical reactors, filtration systems, heat exchangers, and catalytic converters.^{3–7} The complex, tortuous pathways formed by randomly packed particles introduce significant challenges in modeling pore-scale hydrodynamics, making precise predictions difficult.^{8,9}

A cornerstone for predicting pressure drop in such systems has been the semi-empirical Ergun equation,¹⁰ which linearly superposes viscous and inertial losses. While widely used, the Ergun equation and its derivatives, such as the Kozeny–Carman^{11,12} and Burke–Plummer equations, possess inherent limitations. These models assume idealized conditions—smooth, spherical particles and uniform flow distribution—and do not explicitly account for critical factors such as particle surface roughness, pore geometry variations, or the confining wall

effects.^{13,14} Consequently, they often exhibit significant deviations when applied to real-world systems involving rough, irregular, or poly-disperse particles, or systems with low tube-to-particle diameter ratios.^{15,16}

In parallel, the analysis of flow in empty conduits has been dominated by the Reynolds number, a dimensionless quantity that conventionally categorizes flow into laminar, transitional, and turbulent regimes. However, this framework struggles to provide a unified description that seamlessly bridges the flow behavior in empty pipes and particle-packed conduits. The seminal work of Nikuradse¹⁷ on rough pipes highlighted the profound influence of wall effects, yet a fundamental model integrating these findings with packed-bed hydrodynamics remained elusive.

To address these limitations, Quinn¹⁸ recently proposed the Quinn Fluid Flow Model (QFFM), a novel theoretical framework derived from first principles. The QFFM establishes a universal relationship between the normalized dimensionless pressure gradient (P_Q)

and fluid current (C_Q), expressed as $P_Q = k_1 + k_2 C_Q$. A key innovation of the QFFM is its treatment of any closed conduit as a “packed conduit,” whether it contains solid particles or is empty. This is achieved through the conceptual framework of the Hypothetical Q-Channel (HQC) and the introduction of a wall normalization coefficient (λ), which inherently accounts for tortuosity, viscous boundary layers, and surface roughness effects.

Recent experimental and numerical studies have robustly validated the QFFM’s predictive capabilities. Nikitin *et al.*¹⁹ demonstrated excellent agreement between QFFM predictions and data across laminar, transient, and turbulent flow regimes for both water and air, outperforming the Ergun equation. In a subsequent study, Papkov *et al.*²⁰ showed that the QFFM accurately captures the significant impact of particle roughness on pressure drop in randomly packed beds of spherical, cylindrical, and irregular particles, a feat challenging for traditional models.

Despite this successful validation, the full mathematical and physical derivation of the QFFM—spanning its ten sequential development steps from fundamental concepts to the final universal law—has not yet been presented in a standalone, detailed publication. A comprehensive derivation is crucial for the broader acceptance, application, and further development of the model within the fluid dynamics community.

Therefore, this paper aims to present a complete and detailed derivation of the Quinn Fluid Flow Model from first principles. A systematic elaboration on the model’s foundational concepts, including the Hypothetical Q-Channel, the definitions of conduit porosity, and the wall normalization coefficient, is clarified. By providing this thorough theoretical foundation, we seek to bridge the historical gap between the empirical correlations for packed beds and the classical fluid dynamics of empty conduits, offering a unified and physics-based tool for engineers and researchers.

Recently, the authors published two papers with a brief overview of the QFFM.^{15,19} While our two previously published papers in *Physics of Fluids* have successfully demonstrated the QFFM’s predictive capabilities through extensive experimental validation and numerical simulation, they necessarily presented the model’s theoretical foundations in a condensed form due to space constraints. Those works focused primarily on validating the model’s outcomes across various flow regimes and particle morphologies. This paper represents the logical continuation and completion of this research trilogy by providing the comprehensive theoretical derivation that underpins the QFFM. Herein, we present a detailed, step-by-step exposition of the model’s mathematical development from first principles, elaborating on the physical significance of each parameter and the fundamental concepts that were briefly introduced in our previous publications. By providing this thorough theoretical foundation, we complete the presentation of the QFFM and enable its broader adoption and development within the fluid dynamics community.

II. FUNDAMENTALS OF THE QUINN FLUID FLOW MODEL (QFFM)

The Quinn Fluid Flow Model (QFFM) is a theoretical framework derived from first principles to describe pressure-driven fluid flow in packed beds. Its foundation rests on a rigorous reexamination of the structural anatomy of fluid conduits and the application of classical mechanics to fluid motion within a novel conceptual construct: the Hypothetical Q-Channel (HQC). The schematic representation of

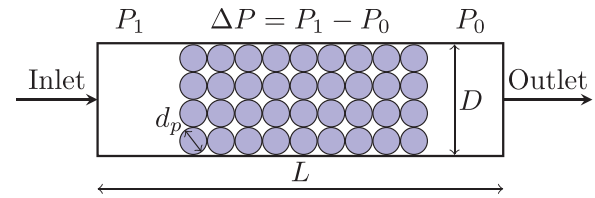


FIG. 1. Fluid flow through a packed bed conduit.

fluid flow through the packed bed is presented in Fig. 1, where particles are defined by diameter d_p , porosity ε , and roughness k .

A. Structural anatomy of a fluid conduit in the QFFM

The QFFM begins by defining the fundamental elements of a fluid flow system with precise, quantitative descriptors. A conduit is defined as a right circular cylinder of length L and diameter D , giving an empty conduit volume of $V_{ec} = \pi D^2 L / 4$.

Within this conduit, particles are treated as spheroidal objects. A critical distinction is made between a particle’s measured diameter, d_{pm} , and its fundamental *spherical particle diameter equivalent*, d_p , which is the diameter of a sphere occupying the same volume as the particle. This relationship is given by $v_{dp} = \pi d_p^3 / 6$, where v_{dp} is the particle volume, determinable via methods like pycnometry. The often-used “sphericity,” Ψ_p , is treated in the QFFM as a correlation factor ($d_p = d_{pm} \Psi_p$) rather than a fundamental property, as d_p is the dimension dictated by the particle’s volume.

Key particle properties include its surface area, $(SA)_p = \pi d_p^2$, and its cross-sectional area, $(CSA)_p = \pi d_p^2 / 4$. From these, the *hydraulic radius*, r_h , is defined as the ratio of surface area to cross-sectional area

$$r_h = \frac{(SA)_p}{(CSA)_p} = 4. \quad (1)$$

This constant is fundamental to the model’s description of fluid–solid interaction. The volume of fluid in the boundary layer when the fluid is at rest is theorized as $V_{bl} = 4\pi r_h^3 / 3$, leading to the definition of the *internal fluid structural constant*

$$k_1 = \frac{V_{bl}}{r_h} \approx 67. \quad (2)$$

This constant, k_1 , represents the normalized viscous control element of the fluid flow.

A conduit is considered “packed” if it contains n_p particles. The variable $n_{pq} = V_{ec} / V_{dp}$ represents the number of particle-equivalents that would fill the entire conduit volume, serving as a scaling factor.

B. Porosity framework and the hypothetical Q-channel (HQC)

The QFFM introduces a comprehensive and hierarchical porosity framework to describe the distribution of solid and fluid phases within a conduit. This framework is essential for unifying the description of both packed and empty conduits.

- Conduit external porosity (ε_0): The fraction of the conduit volume external to the particles, defined as $\varepsilon_0 = V_e / V_{ec}$, where V_e is the volume between particles.

- Particle volume fraction ($1 - \varepsilon_0$): The fraction occupied by the particles themselves, $(1 - \varepsilon_0) = n_p/n_{pq} = V_p/V_{ec}$.
- Particle porosity (ε_p): The porosity internal to the particles ($\varepsilon_p = V_i/V_p$).
- Conduit total porosity (ε_t): The total fraction of the conduit volume available to the fluid, $\varepsilon_t = \varepsilon_0 + \varepsilon_i$, where ε_i is the volume fraction of intra-particle pores.

The cornerstone of the QFFM is the Hypothetical Q-Channel (HQC), a normalized unit conduit derived from this porosity framework. The characteristic diameter of the HQC is defined as

$$d_c = \frac{d_p}{\text{abs}(1 - \varepsilon_0)}. \quad (3)$$

This definition leads to three critical limits that define the HQC's behavior across the porosity continuum:

- Conduit packed with solid matter: As the number of solid particles approaches the maximum ($n_p \rightarrow n_{pq}$), the external porosity approaches zero ($\varepsilon_0 \rightarrow 0$) and the HQC diameter collapses to zero ($d_c \rightarrow 0$).
- Conduit with no particles: In the limit of a particle-free conduit ($n_p \rightarrow 0$), the external porosity approaches unity ($\varepsilon_0 \rightarrow 1.0$) and the HQC diameter becomes infinite ($d_c \rightarrow \infty$). It is crucial to note that a value of $\varepsilon_0 = 1.0$ is not permitted under the QFFM doctrine, as it represents a mathematical discontinuity not permitted. This specific condition, where $d_c = d_p/0 \rightarrow \infty$, reveals a fundamental limitation in conventional fluid dynamics.
- Conduit packed with “free space” particles (empty conduit): For an empty conduit, the model defines $n_p \rightarrow -n_{pq}$, leading to $\varepsilon_0 \rightarrow 2$ and, critically, $d_c \rightarrow D$. This establishes the empty conduit not as a special case with separate physics, but as a specific boundary condition within the general QFFM framework.

Thus, the empty conduit is not a special case with its own separate physics, but rather a specific boundary condition within the general QFFM framework. The cross-sectional area and length of the unit HQC are $A_c = \pi d_c^2/4$ and $L_c = 1/A_c$, respectively, ensuring a normalized volume of unity.

C. Fluid motion as curvilinear simple harmonic motion

The QFFM posits that the motion of a fluid element within the HQC follows the principles of Curvilinear Simple Harmonic Motion (SHM) as schematically presented in Fig. 2. This is a radical departure from conventional approaches and provides a unified kinematic description.

In SHM, the phase of the motion, Φ , is given by

$$\Phi = \omega t + \alpha, \quad (4)$$

where α is the initial phase (epoch), ω is the angular velocity, and t is the time. The QFFM maps these mechanical terms directly onto fluid dynamic parameters:

- Phase, $\Phi \rightarrow P_Q$, the normalized dimensionless pressure gradient.
- Initial phase, $\alpha \rightarrow k_1 \approx 67$, the universal viscous constant.
- Angular velocity, $\omega \rightarrow k_2 \lambda$, where $k_2 \approx 0.04$ is the universal kinetic constant and λ is the wall normalization coefficient.
- Time, $t \rightarrow Q_N = \delta R_{em}$, the Quinn number.

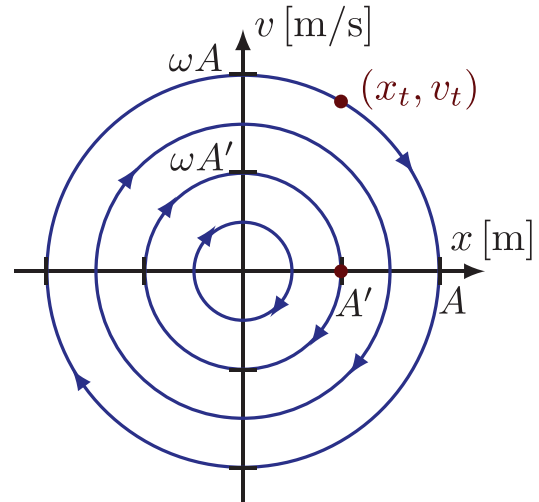


FIG. 2. Phase portrait of a simple harmonic motion in Q-channel: (x_t, v_t) is the position at any time t of a point describing a circle of radius a with uniform angular velocity ω about the center O .

This mapping yields the fundamental equation of the QFFM, known as Quinn's law

$$P_Q = k_1 + k_2 C_Q, \quad (5)$$

where $C_Q = \lambda \delta R_{em}$. This linear relationship between the normalized pressure gradient (P_Q) and the fluid current (C_Q) is the central tenet of the model. The position, instantaneous velocity, and acceleration of a fluid element within the HQC can be described by the standard equations of SHM, with the phase defined by Quinn's Law. The period of this motion is a constant, $T = 16\pi^2 \approx 158$, representing a fundamental time constant of the system.

D. Wall normalization coefficient (λ)

The wall normalization coefficient, λ , is a dimensionless parameter that quantifies the total additional flow resistance arising from the presence of solid boundaries and their interaction with the fluid. Its value is determined by the net wall effect, W_N , which is the sum of two distinct mechanisms:

1. Primary wall effect (W_1): This is the flow resistance due to the viscous boundary layer that develops along confining walls. It is a function of the conduit architecture (tortuosity, τ), fluid flow rate, and fluid viscosity.
2. Residual secondary wall effect (W_{2R}): This represents the resistance due to surface roughness, but only when the roughness elements “punch through” the viscous boundary layer. The model includes a conditional adjustment such that roughness only impacts the flow profile when its effective height exceeds the boundary-layer thickness.

In summary, the net wall-effect can be expressed as

$$W_N = W_1 + W_{2R}. \quad (6)$$

Consequently, the wall normalization coefficient can be defined as

$$\lambda = 1 + W_N. \quad (7)$$

A value of $\lambda = 1$ indicates a theoretically ideal packed bed with no wall effects. Values of $\lambda > 1$ signify increased flow resistance due to the primary effect (dominant in smooth-walled systems), the secondary effect (dominant in rough-walled systems), or a combination of both. This coefficient is the vehicle through which the QFFM incorporates the seminal findings of Nikuradse¹⁷ on rough pipes into a unified framework for all closed conduits.

III. DETAILED DERIVATION OF THE QUINN FLUID FLOW MODEL

The derivation of the QFFM proceeds through a sequence of logical steps, beginning with a general expression for pressure drop and culminating in the universal linear relationship of Quinn's Law. This process systematically introduces the model's core concepts and demonstrates how they are integrated from first principles.

A. Step 1: General form of the pressure-flow relationship

The starting point is a generalized equation for pressure drop in a porous medium, which is a second-order polynomial with respect to flow rate, capturing both viscous and inertial losses:

$$\frac{\Delta P}{L} = aq + bq^2, \quad (8)$$

where a and b are empirical coefficients that balance the equation for a specific system.

B. Step 2: Forchheimer equation form

Expressing this relationship in terms of the hydraulic gradient and superficial velocity (μ_s) yields a form analogous to the Forchheimer equation

$$\frac{\Delta P}{\rho_f g L} = a_{For} \mu_s + b_{For} \mu_s^2, \quad (9)$$

where a_{For} and b_{For} are the Forchheimer coefficients, ρ_f is the fluid density, and g is the acceleration due to gravity.

C. Step 3: Substitution with QFFM parameters

The QFFM replaces the empirical Forchheimer coefficients with physically meaningful parameters derived from the Hypothetical Q-Channel (HQC). The coefficients are substituted as follows:

$$\frac{\Delta P}{\rho_f g L} = \frac{4r_h^2 \pi \delta \eta \mu_s}{3 \rho_f g d_c^2} + \frac{\delta^2 \lambda \mu_s^2}{2 \pi g d_c}. \quad (10)$$

Here, the fundamental QFFM parameters are introduced:

- $r_h = 4$, the hydraulic radius.
- $\delta = 1/\varepsilon_0^3$, the porosity normalization coefficient.
- η , the fluid absolute viscosity.
- $d_c = d_p / \text{abs}(1 - \varepsilon_0)$, the HQC diameter.
- λ , the wall normalization coefficient.

D. Step 4: Rearrangement and simplification

Multiplying both sides of the equation by $\rho_f g$ simplifies the expression and returns it to a pressure gradient form

$$\frac{\Delta P}{L} = \frac{268 \delta \eta \mu_s}{d_c^2} + \frac{\delta^2 \lambda \rho_f \mu_s^2}{2 \pi d_c}. \quad (11)$$

The constant 268 arises from the evaluation $(4r_h^2 \pi)/3$ with $r_h = 4$, giving $(4 \cdot 16 \cdot \pi)/3 \approx 268$.

E. Step 5: Introduction of hydraulic forces

To group terms physically, the model defines two key hydraulic forces per unit control volume:

- Viscous hydraulic force: $n_v = \frac{\delta \eta \mu_s}{d_c^2}$
- Kinetic hydraulic force: $n_k = \frac{\delta \rho_f \mu_s^2}{d_c}$

Substituting these definitions yields

$$\frac{\Delta P}{L} = 268 n_v + \frac{\delta \lambda}{2 \pi} n_k. \quad (12)$$

F. Step 6: Normalization for viscous effects

To non-dimensionalize the equation, both sides are divided by the viscous hydraulic force, n_v

$$\frac{\Delta P}{L n_v} = 268 + \frac{\delta \lambda n_k}{2 \pi n_v}. \quad (13)$$

G. Step 7: Q-modified Ergun equation

The left-hand side is defined as the viscous friction factor, $f_v = \Delta P / (L n_v)$. The ratio n_k / n_v is identified as a modified Reynolds number, Re_m . The coefficient of Re_m is defined as the kinetic constant B .

This gives the Q-modified Ergun equation

$$f_v = A + B Re_m, \quad (14)$$

where

- $f_v = \frac{\Delta P}{L n_v}$, viscous friction factor.
- $A = 268$, viscous constant.
- $B = \frac{\delta \lambda}{2 \pi} = \frac{\lambda}{2 \pi \varepsilon_0^3}$, kinetic constant.
- $Re_m = \frac{n_k}{n_v}$, modified Reynolds number.

Critically, unlike the classical Ergun equation, where $B = 1.75$, the QFFM defines B as a variable function of the wall normalization coefficient (λ) and the external porosity (ε_0).

H. Step 8: Normalization for hydrodynamic drag

To incorporate the universal constants k_1 and k_2 , the equation is normalized by the fluid drag normalization coefficient, $r_h = 4$

$$\frac{f_v}{4} = 67 + \frac{\delta \lambda Re_m}{8 \pi}. \quad (15)$$

I. Step 9: Quinn's law—The universal flow equation

The terms in the equation from Step 8 are now identified as the core dimensionless groups of the QFFM:

- Normalized pressure gradient: $P_Q = \frac{f_c}{4}$
- Universal viscous constant: $k_1 = 67$
- Universal kinetic constant: $k_2 = \frac{1}{8\pi} \approx 0.04$
- Fluid current: $C_Q = \delta \lambda R e_m$

This leads to the fundamental linear relationship of the QFFM, known as Quinn's law

$$P_Q = k_1 + k_2 C_Q. \quad (16)$$

This equation is universal, applicable from creeping to fully turbulent flow in both packed and empty conduits.

J. Step 10: Dimensionless permeability and the Quinn number

Finally, the relationship is rearranged to define the dimensionless permeability (Θ) of the HQC as a function of the instantaneous fluid current (Q_N), which is synonymous with C_Q .

Starting from Quinn's law

$$P_Q = k_1 + k_2 Q_N. \quad (17)$$

The dimensionless permeability is defined as the ratio of fluid current to pressure gradient

$$\Theta = \frac{Q_N}{P_Q} = \frac{Q_N}{k_1 + k_2 Q_N}. \quad (18)$$

This can be rearranged to highlight the role of the wall normalization coefficient, λ

$$\Theta = \frac{1}{\frac{k_1}{Q_N} + \frac{\lambda}{2\pi r_h}}. \quad (19)$$

This expression demonstrates that the permeability is inversely related to the wall normalization coefficient λ , quantitatively capturing how wall effects and surface roughness reduce the effective permeability of the conduit.

The physical significance of λ and its role in quantifying the transition of flow regimes is powerfully illustrated by plotting the dimensionless permeability Θ as a function of the instantaneous fluid current Q_N as shown in Fig. 3. This representation provides an intuitive visualization of the fluid dynamic processes occurring as wall roughness increases. As the sand-roughness coefficient k , and consequently the relative roughness k/d_c , increases, the wall normalization coefficient λ also rises. The Θ vs Q_N plot vividly demonstrates how this increasing effective roughness “punches through” the viscous boundary layer at progressively lower values of Q_N . This occurs because a higher λ signifies a more dissipated boundary layer, allowing the roughness elements to exert their influence on the core flow earlier, i.e., at a lower fluid current. This graphical relationship consolidates the complex interplay between viscous forces, inertial forces, and surface roughness into an intuitive, coherent picture. The plot makes it immediately apparent how increased wall resistance, captured by λ ,

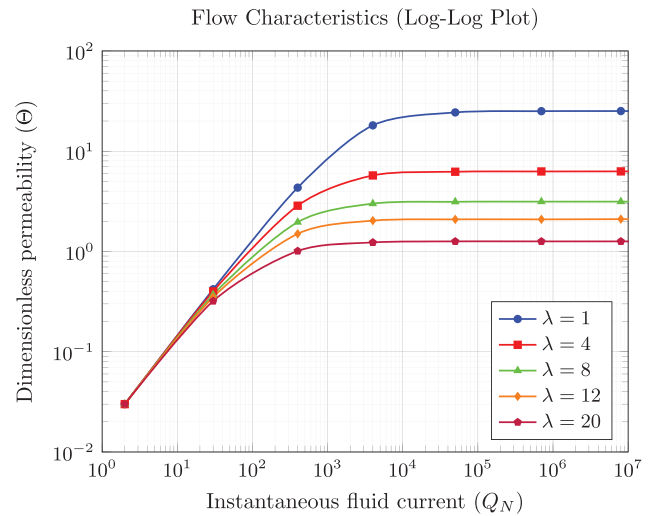


FIG. 3. Log-log plot of dimensionless permeability (Θ) vs instantaneous fluid current (Q_N) for different values of λ .

systematically reduces the system's dimensionless permeability across the flow regime.

K. Final pressure drop equation

By working backward through the derivation and substituting the definitions of n_v , n_k , d_c , and δ , the final expression for predicting pressure drop in a packed bed is obtained

$$\frac{\Delta P}{L} = \frac{A(1 - \varepsilon_0)^2 \mu_s \eta}{\varepsilon_0^3 d_p^2} + \frac{B(1 - \varepsilon_0) \rho_f \mu_s^2}{\varepsilon_0^3 d_p}. \quad (20)$$

While this maintains the general structure of the Ergun equation, the constants are now fundamentally defined: $A = 268$ and $B = \lambda / (2\pi \varepsilon_0^3)$. This final equation integrates the effects of particle size (d_p), bed porosity (ε_0), fluid properties (η , ρ_f), and the comprehensive wall effect (λ) into a single, predictive model.

This ten-step derivation establishes the QFFM not as an empirical correlation, but as a physics-based framework rooted in the anatomy of the flow conduit and the principles of simple harmonic motion.

IV. VALIDATION OF THE QUINN FLUID FLOW MODEL

The predictive capability and robustness of the Quinn Fluid Flow Model have been extensively validated through both experimental investigations and high-fidelity computational fluid dynamics (CFD) simulations across diverse flow regimes, particle morphologies, and operational conditions. This section summarizes the key validation results from previous studies that demonstrate the model's accuracy and universality.

A. Experimental validation with smooth spherical particles

Initial validation of the QFFM was conducted using packed beds of polished spherical particles under controlled laboratory conditions.¹⁹ The experimental setup consisted of a recirculating flow loop with precise pressure and temperature measurement capabilities. A packed column

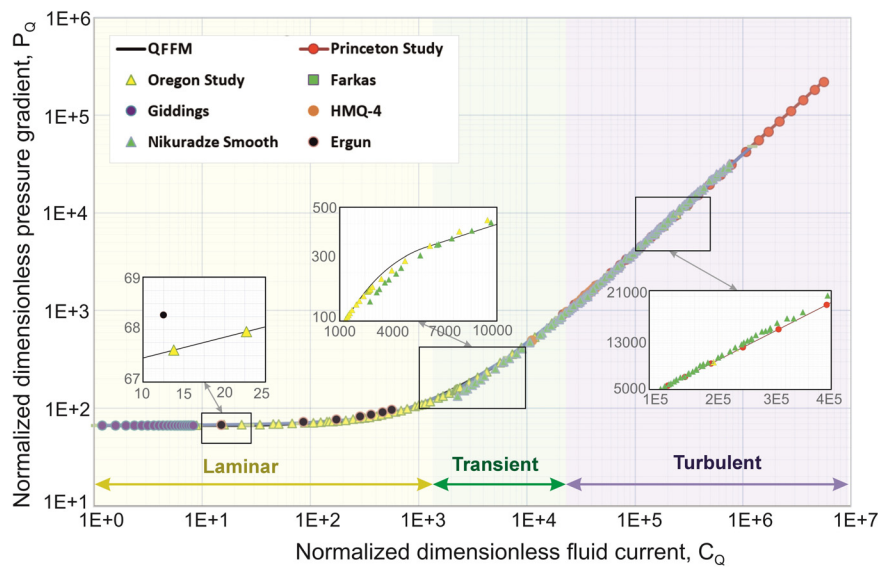


FIG. 4. Dimensionless log–log plot (insets are linear–linear) of viscous friction factor P_Q vs wall-effect normalized fluid current C_Q according to the different sources: Oregon study;²¹ Giddings;²² Nikuradse Smooth;¹⁷ Princeton Study;^{23,24} Farkas *et al.*;²⁵ Ergun;¹⁰ HMQ-4 – authors’ experimental data. Reproduced from Nikitin *et al.*, “Flow through packed beds: Experimental and numerical validation of the Quinn fluid flow model,” *Phys. Fluids* 37, 084113 (2025) with the permission of AIP Publishing.

with an inner diameter $D = 3.85$ mm and length $L = 1032.67$ mm was filled with 297 electro-polished stainless steel spherical particles ($d_p = 3.469$ mm $\pm 0.5\%$), yielding a bed porosity of $\varepsilon_0 = 0.4557$. The experimental data in Fig. 4 are designated as HMQ-14.

The QFFM solver was applied to this configuration, determining that the spherical particles were hydrodynamically smooth with a sand roughness coefficient $k = 0$. The wall normalization coefficient was found to be $\lambda \approx 1.003$, indicating negligible secondary wall effects with only a minimal primary wall effect due to the viscous boundary layer. Comparison between measured pressure drops and QFFM predictions revealed exceptional agreement across the entire flow range, with a mean absolute error (MAE) of approximately 1%.

To assess the performance of the Quinn Fluid Flow Model (QFFM), the correlation between the normalized dimensionless pressure gradient (P_Q) and fluid current (C_Q) was examined using a logarithmic scale. Experimental data were evaluated alongside findings from several independent investigations, including pipe flow research from Oregon²¹ and Princeton Universities,^{23,24} packed-bed measurements by Giddings,²² smooth-pipe data from Nikuradse,¹⁷ the validation work of Farkas *et al.*,²⁵ and the foundational fluidization studies of Ergun.¹⁰

As illustrated in Fig. 4, the compiled dataset, representing a wide range of flow conditions from laminar to turbulent regimes, aligns closely with the predictions of the QFFM. The log–log plot demonstrates the model’s consistent accuracy across this broad spectrum of operational states.

The MAE metric was preferred over root mean squared error (RMSE) as it does not exaggerate large errors, which is critical when errors are not normalized. The excellent correlation confirmed the QFFM’s legitimacy for smooth spherical particle systems and demonstrated its ability to accurately back-calculate key parameters such as external porosity (ε_0) when particle diameter (d_p) is known.

B. Numerical validation with structured packings

Complementary to experimental validation, comprehensive CFD simulations were performed to validate the QFFM across different

geometric configurations and fluid types.¹⁹ Two distinct computational domains were developed:

1. Cylindrical domain (water flow)

A cylindrical domain ($D = 3.85$ mm, $L = 115$ mm) containing 30 spherical particles ($d_p = 3.469$ mm) arranged in a linear array was simulated for water flow ($\rho_f = 1000$ kg/m³, $\eta = 1$ mPas). Mesh sensitivity analysis confirmed convergence at cell sizes ≤ 0.25 mm with error $< 1\%$. The QFFM predictions showed excellent agreement with CFD results, with root mean square discrepancy below 1% and maximum absolute error of 3.7% across the flow rate range up to 0.56 g/s.

2. Square domain (air flow)

A square domain (20×20 mm cross section, $L = 260$ mm) packed with 64 spherical particles ($d_p = 10.0001$ mm) in orthogonal array configuration was simulated for air flow ($\rho_f = 1.194$ kg/m³, $\eta = 1.81e-5$ Pas). The QFFM matched experimental measurements from Bai’s paper²⁶ with an average deviation of 0.55 kPa (2.2% of full scale) across flow rates $G^m = 2.892$ to 7.342 g/s.

The numerical studies revealed that the QFFM outperformed the traditional Ergun equation, particularly for low D/d_p ratios where the Ergun equation exhibited deviations up to 15%. The model’s inherent accounting for tortuosity and microscale flow heterogeneity proved essential for accurate predictions in confined geometries.

C. Validation with rough and irregular particles

The QFFM’s capability to handle non-ideal particle systems was rigorously tested using rough cylindrical particles and irregularly shaped clinker particles.²⁰ This investigation was crucial for demonstrating the model’s superiority over traditional correlations that assume smooth, spherical particles.

Figure 5(a) illustrates the relationship between dimensionless permeability (Θ) and instantaneous fluid current (Q_N) for cylindrical

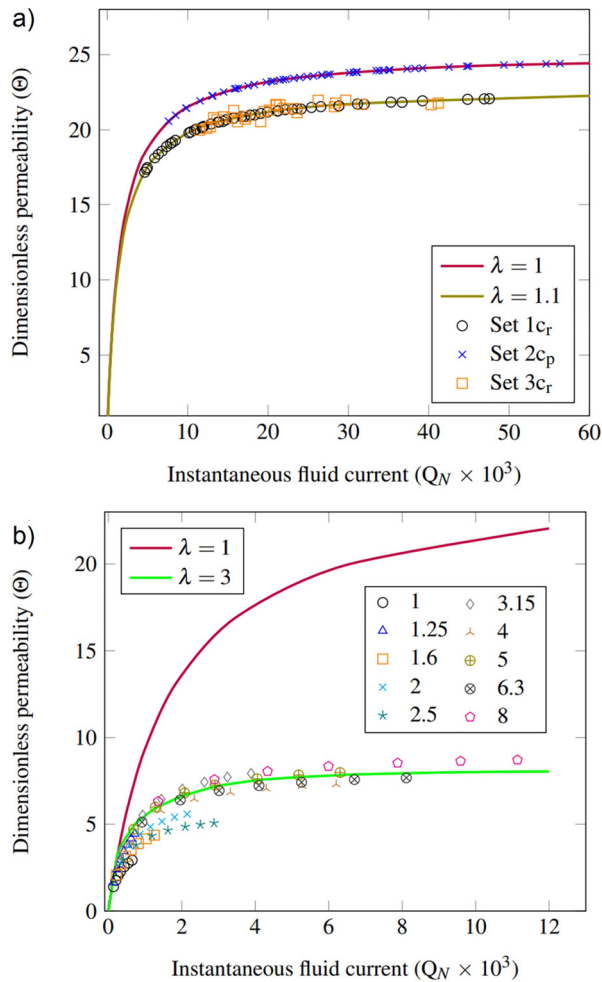


FIG. 5. Dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for or various systems: (a) packed bed of cylindrical particles: line—predicted data via QFFM, points—measured data²⁰ for Set 1c_r, Set 2c_p, Set 3c_r; (b) packed bed of irregularly shaped particles: line—predicted data via QFFM, points—measured data from Ref. 27. Partially reproduced from Papkov *et al.*, “Flow through randomly packed beds of rough particles: Model prediction and experimental validation,” *Phys. Fluids* 37, 103318 (2025) with the permission of AIP Publishing.

particles,^{15,20} comparing experimental data with predictions from the Quinn Fluid Flow Model (QFFM). The results show excellent agreement between measured and modeled data for polished cylindrical particles, which align closely with the theoretical line for smooth particles ($\lambda \approx 1$). In contrast, rough cylindrical particles ($\lambda \approx 1.1$) exhibit a clear reduction in dimensionless permeability, confirming that surface roughness increases flow resistance and elevates pressure drop in packed beds.

Figure 5(b) extends this analysis to irregularly shaped clinker particles from Buchwald’s paper,²⁷ demonstrating a more pronounced effect of surface roughness on permeability. The wall normalization coefficient (λ) for these particles exceeds 3, leading to a substantial decrease in dimensionless permeability compared to hypothetical smooth particles under equivalent conditions. This

significant reduction underscores the limitations of traditional models like Ergun or Kozeny–Carman for rough or irregular particles and highlights the QFFM’s ability to accurately capture the impact of surface roughness across diverse particle morphologies and flow regimes.

1. Rough cylindrical particles

Experiments were conducted with both rough and polished cylindrical particles in packed beds with diameter-to-particle ratios $D/d_p = 4.64$ and 7.68 . The QFFM solver successfully quantified the significant effect of particle roughness on the wall normalization coefficient. For polished particles, $\lambda \approx 1$, while for rough particles, λ increased to approximately 1.1. This 10% increase in λ directly influenced the kinetic constant B in the Q-modified Ergun equation, leading to a 20%–30% reduction in dimensionless permeability (Θ) compared to smooth particles.

The experimental data and QFFM predictions, when expressed in dimensionless form (Θ vs Q_N), clearly demonstrated how particle roughness affects pressure drop in packed beds. The results showed excellent agreement between measured and modeled data, confirming the QFFM’s ability to capture roughness effects through the λ parameter.

2. Irregular clinker particles

To evaluate the model’s performance with highly irregular particles, experimental data for clinker particles from Ref. 27 were analyzed using the QFFM framework. The clinker particles exhibited significant surface roughness, with wall normalization coefficients λ exceeding 3–6 due to their high surface roughness ($k \sim 10^{-4}$ – 10^{-5} cm).

The QFFM analysis revealed that these rough, irregular particles exhibited permeability reductions of 50%–70% relative to smooth-particle beds. The dimensionless representation of permeability vs fluid current enabled systematic evaluation of packed bed performance across various operational and design parameters, demonstrating the QFFM’s ability to handle extreme surface roughness and irregular geometries that challenge traditional models.

D. Unification of packed bed and empty conduit flow

A particularly powerful demonstration of the QFFM’s universality is its ability to seamlessly bridge the gap between packed bed flow and empty conduit flow within a single theoretical framework. As shown in Fig. 4, the QFFM naturally encompasses both regimes through appropriate parameter limits.

For empty conduits, the QFFM parameters approach specific limits:

- Particle porosity: $\varepsilon_p \rightarrow 1$ (particles are entirely void space)
- Conduit total porosity: $\varepsilon_t = 1$ (entire volume available to fluid)
- Characteristic conduit diameter: $d_c \rightarrow D$ (hydraulic diameter becomes pipe diameter)
- Wall normalization coefficient: $\lambda > 1$ (determined by primary and residual secondary wall effects)

In this limit, the QFFM equations reduce to the governing equations for flow in circular pipes, and the dimensionless permeability Θ follows the characteristic curve of pipe flow, including the distinct

laminar-to-turbulent transition. The model successfully reproduces the classical Moody chart behavior²⁸ and Nikuradse's rough pipe data,¹⁷ demonstrating how it naturally encompasses both packed beds and empty conduits within a unified theoretical framework.

E. Performance across flow regimes

The QFFM has been validated across the complete spectrum of flow regimes, from creeping flow to fully turbulent conditions. The model's linear relationship $P_Q = k_1 + k_2 C_Q$ remains valid throughout these regimes, with the relative contributions of the viscous (k_1) and kinetic ($k_2 C_Q$) terms naturally adjusting to match the physical behavior.

In laminar flow regimes, the kinetic term becomes negligible, and the QFFM reduces to a form analogous to the Kozeny–Carman equation. As flow transitions to turbulent conditions, the kinetic term dominates, reproducing the behavior described by the Burke–Plummer equation. However, unlike these traditional models that are regime-specific, the QFFM provides continuous, accurate predictions across all regimes without requiring empirical switching functions or regime-based correlations.

The validation studies confirm that the QFFM's dimensionless analysis successfully differentiates between primary (viscous boundary layer) and secondary (roughness-induced) wall effects, providing physical insight into the fundamental mechanisms governing pressure drop in porous media.

F. Summary of validation evidence

The comprehensive validation of the QFFM across multiple studies demonstrates:

- Accuracy: MAE $\approx 1\%$ for smooth spherical particles; deviations $< 3.7\%$ across various configurations
- Robustness: Consistent performance across laminar, transitional, and turbulent flow regimes
- Universality: Applicable to both packed beds and empty conduits within a single framework
- Physical Basis: Successful quantification of roughness effects through the wall normalization coefficient λ
- Predictive capability: Ability to back-calculate difficult-to-measure parameters (ε_0 , λ , d_p) from limited experimental data
- Superiority over traditional models: Outperforms the Ergun equation, particularly for rough particles, irregular geometries, and low D/d_p ratios

This extensive validation establishes the QFFM as a reliable, physics-based tool for predicting pressure drop in closed conduits, offering significant advantages over traditional empirical correlations for both academic research and industrial applications.

V. CONCLUSION

This paper has presented a comprehensive derivation and theoretical foundation of the Quinn Fluid Flow Model, establishing it as a unified framework for predicting pressure-driven fluid flow in closed conduits. Through systematic development from first principles, we have demonstrated that the QFFM successfully bridges the historical gap between traditional packed-bed correlations and empty-conduit

flow models, offering a physics-based alternative to semi-empirical approaches.

The model's fundamental innovation lies in its treatment of all closed conduits as packed systems through the Hypothetical Q-Channel concept, which provides a normalized framework for analyzing fluid flow across the entire porosity continuum. The derivation of Quinn's Law reveals a universal linear relationship between the normalized pressure gradient and the fluid current, with universal constants derived from geometric and physical principles rather than empirical fitting. The wall normalization coefficient quantitatively captures the combined effects of viscous boundary layers and surface roughness, providing physical insight into the mechanisms governing energy dissipation in confined flows.

The QFFM fundamentally advances fluid dynamics modeling by addressing critical limitations of traditional approaches. Unlike the Ergun equation, which treats its coefficients as empirical constants, the QFFM derives its parameters from first principles and explicitly accounts for porosity effects and wall interactions. In contrast to Reynolds number-based correlations, the QFFM introduces the Quinn number as the governing parameter for flow regime characterization, incorporating the essential elements of porosity and wall effects that are missing from conventional analyses. The model provides continuous predictions from creeping to fully turbulent flow without requiring regime-specific correlations or empirical switching functions, demonstrating true universality across the entire flow spectrum.

Extensive validation has demonstrated the model's exceptional predictive capability across diverse systems and conditions. The QFFM achieves remarkable accuracy for smooth spherical particles, accurately captures the significant effects of surface roughness, and successfully handles irregular particle geometries that challenge traditional correlations. The solver capability represents a practical advancement for industrial applications, enabling the determination of difficult-to-measure parameters from basic experimental data.

The theoretical framework presented here provides a rigorous foundation for future extensions to more complex systems, including polydisperse packings, multiphase flows, reactive systems, and non-Newtonian fluids. The integration of the QFFM with computational tools for industrial reactor design and optimization represents a promising direction for practical implementation. By unifying packed-bed and empty-conduit flow within a single theoretical framework and deriving its parameters from first principles, the Quinn Fluid Flow Model establishes a new paradigm in fluid dynamics that offers both fundamental insight and practical utility for engineering applications.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

H. M. Quinn: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **D. Pashchenko:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation

(equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

NOMENCLATURE

Abbreviations

HQC	Hypothetical Q Channel
MAE	Mean Absolute Error
QFFM	Quinn Fluid Flow Model
RMSE	Root Mean Squared Error

Greek symbols

δ	Porosity normalization coefficient
γ_a	Area-weighted uniformity index
λ	Wall normalization coefficient
μ_s	Fluid superficial velocity
ν	Fluid absolute viscosity
ρ_f	Fluid density
Θ	Dimensionless permeability
ε	Q-Porosity Function
ε_0	Conduit external volume fraction
ε_i	Conduit internal volume fraction
ε_p	Particle porosity
ε_t	Conduit total volume fraction
ε_{sk}	Conduit particle skeletal volume fraction

Roman symbols

A	Viscous constant
A_h	Hydraulic cross-section area
B	Kinetic constant
C_Q	Fluid current for fluid flow in closed conduits
d_c	Hypothetical Q-channel diameter
d_p	Average spherical particle diameter equivalent
f_v	Viscous friction factor
g	Acceleration due to gravity
G^m	Fluid mass flow rate
I	Turbulent intensity
k_1	Universal viscous constant
k_2	Universal kinetic constant
L	Channel length
L_m	Turbulent length scale
n_k	Kinetic hydraulic force
n_p	Number of particles
n_{pq}	Volume of the conduit expressed in terms of number of particles
n_v	Viscous hydraulic force
ΔP	Pressure loss
P_Q	Normalized dimensionless pressure gradient
q	Volumetric flow rate
Q_N	Instantaneous fluid current
R_t	Turbulent viscosity ratio

Re_m	Modified Reynolds number
r_h	Fluid drag normalization coefficient

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